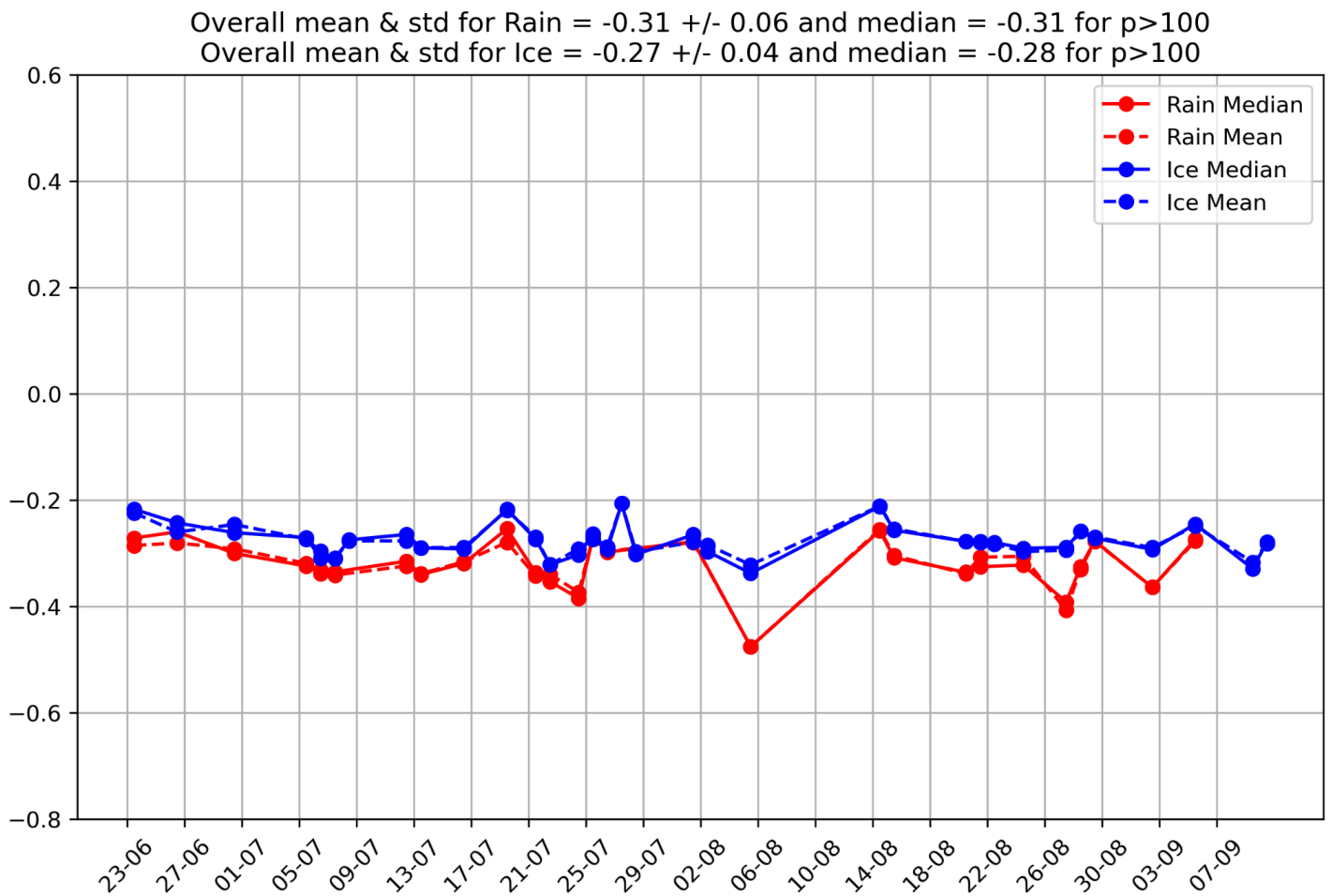
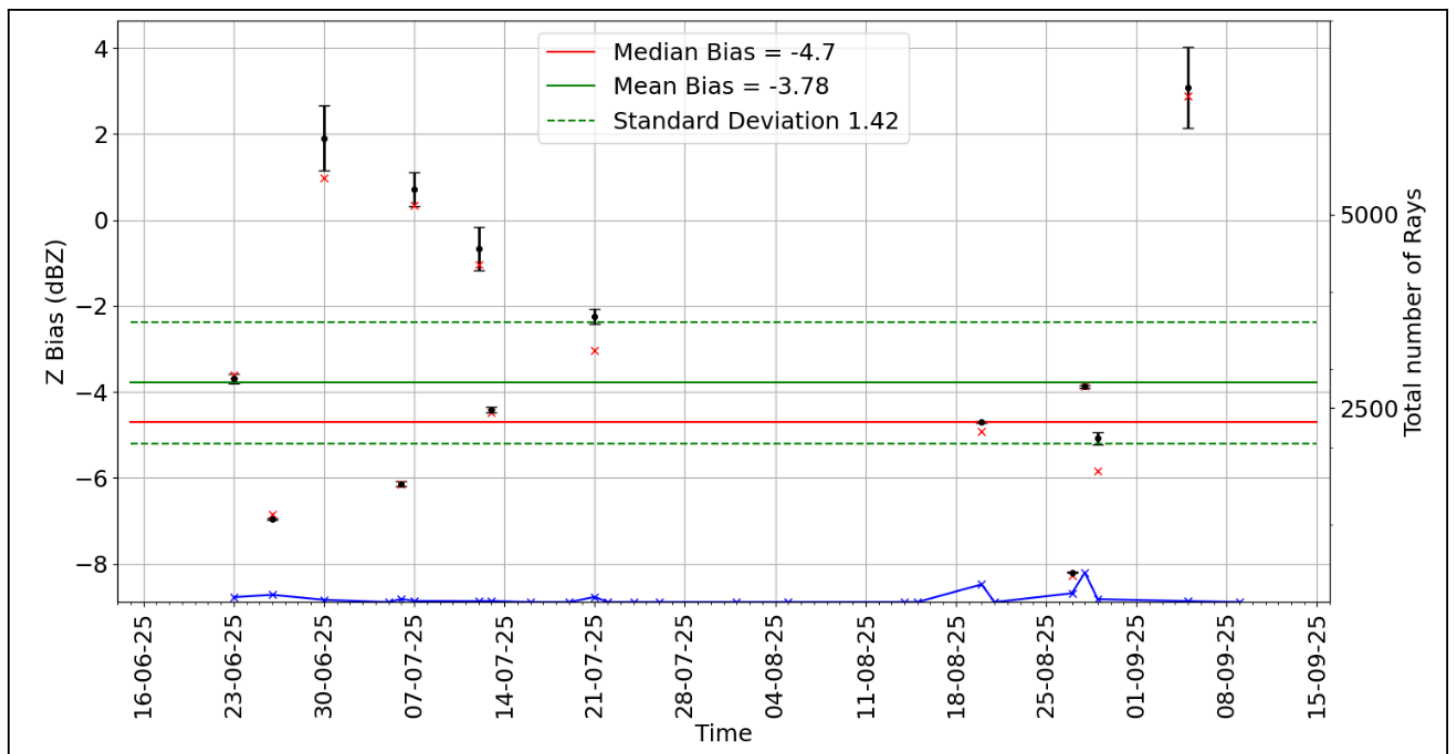


Summary of results: +0.3 offset applied to ZDR and ZDRu, no offset applied to Z

Differential Reflectivity (ZDR) bias estimated from analysis of birdbath scans



Reflectivity bias calculated from the self-consistency analysis



The selection of rays for analysis permitted 2 values of ZDR > 2dB and 2 values of RhoHV < 0.98 in a ray. If there were more, they were excluded. For previous dataset analyses, these criteria were even stricter, but for the TEAMx dataset, very few rays met the criteria, so we relaxed them.

A criteria of a minimum of 10 good rays in a volume was specified, otherwise the volume was rejected. The blue line on the plot above shows the total number of good rays for each day.

20250623	64
20250626	94
20250630	30
20250706	41
20250707	15
20250712	12
20250713	14
20250721	67
20250820	228
20250827	114
20250828	381
20250829	38
20250905	13

The amount of data available for analysis was much lower than previous projects. This is because the radar was located at an altitude of ~2.5km and there was a lack of data below the melting layer that met the criteria for analysis. For the previous project, WOEST, we found a bias of -3.5dBZ (cloud scans) and -3.8dBZ (boundary-layer scans). The plot above suggests the bias for TEAMx is similar, however we are not confident enough to apply this to the dataset, but users should bear this in mind. Future analysis may be undertaken to try to establish a reflectivity bias, and an updated dataset issued.